Pitch deck - dURL

Theme 4: Open Track

Hedera Hello Future Apex Hackathon 2026

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dURL //dev

Live demo: https://durl.dev

GitHub repo: https://github.com/MauriceBoendermaker/hedera_hackathon_2026

Date
21 March 2026

Version
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01 - TEAM & PROJECT INTRODUCTION

Meet the team

We are four developers dedicated to make blockchain-based links as easy as bit.ly

Maurice Boendermaker	Yassine Abderrazik	Mathijs de Niet	Thijs van Steenbeek
Blockchain & Full-Stack Dev	Blockchain & Full-Stack Dev	SDK Developer	SDK Developer

About durl.dev

- Combines Hedera Smart Contract Service for slug storage with HCS for immutable audit logging
- Frontend built with ethers.js + MetaMask, Bootstrap 5, TypeScript
- Live at https://durl.dev

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02 - THE PROBLEM

Links break. Links disappear

Centralised shorteners are a single point of failure for the web

01

Centralised Control

Bit.ly, TinyURL and others can delete, change or block any link at any time.

02

No Transparency

You cannot verify who created a link or when, making phishing trivial

03

Platform Risk

If the service shuts down, every shortened link on the web goes dead.

04

Web3 Gap

No shortener exists that is natively on-chain, auditable, and permanent.

~14B

shortened links created/year

100%

rely on centralised providers

0

fully on-chain shorteners on Hedera.
Until now

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03 - PROJECT SUMMARY

Our solution

Permanent, verifiable short URLs, stored and audited entirely on Hedera

How it works

1. User connects MetaMask wallet to durl.dev
2. Enters a URL > gets a keccak256-generated slug
3. Slug + URL stored on Hedera Smart Contract
4. Creation event logged immutably to HCS topic
5. Anyone can resolve or audit any link, forever

2 link types

1. **Random link** *(free)*
   keccak256(sender + URL + timestamp)
2. **Custom slug** *(HBAR fee)*
   User-chosen slug, reserved on-chain

Hedera Services used

- **Smart Contract Service**
  Slug-to-URL mapping, custom link payment enforcement
- **Hedera Consensus Service**
  Immutable, timestamped audit log of every link creation
- **Mirror Node REST API**
  Reading HCS message history for audit/history pages

04 - INNOVATION

Built different.

durl.dev does something that has never been done on Hedera (or any chain)

Feature	Bit.ly / TinyURL	Other Web3	durl.dev
Permanent links	No	Partial	Yes
Tamper-proof	No	Partial	Yes
Public audit log	No	No	Yes (HCS)
EVM wallet compatible	N/A	Yes	Yes
On Hedera	No	No	Yes
Custom slugs on-chain	No	Rare	Yes

Why Hedera specifically?

- Low, predictable fees make per-link on-chain storage economically viable at scale
- HCS provides ordered, timestamped consensus. Ideal for a public link audit log
- EVM compatibility lets us use MetaMask with no custom wallet needed

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05 - EXECUTION & MVP

What we built

A fully functioning MVP. Migrated, extended, and live.

Random URL Shortening

Connect wallet, paste URL, get a permanent on-chain short link instantly.

Custom Slug Registration

Pay a small HBAR fee to reserve a human-readable slug of your choice.

HCS Audit Log

Every link creation is submitted as a signed message to a public HCS topic.

Public Resolution

Any durl.dev/* link resolves via smart contract. No backend needed.

Mirror Node History

Full creation history readable from the Hedera mirror node REST API.

MetaMask Integration

No new wallet required, it works with any EVM-compatible wallet.

Tech stack

- TypeScript
- Ether.js
- @Hashgraph/sdk
- Solidity
- Bootstrap 5
- Hardhat
- MetaMask

06 - SUCCESS & HEDERA IMPACT

Why this grows Hedera

Every short link created is a Hedera transaction. Every click is an advertisement for the network.

New Hedera Accounts

Every new user who wants to shorten a link must connect an EVM-compatible Hedera account, directly driving account creation.

Higher TPS

Each link creation writes to the smart contract AND submits an HCS message, two transactions per action, compounding network activity.

Mainstream Exposure

URL shorteners are used by everyone. durl.dev is a daily-use utility that introduces non-crypto users to Hedera naturally.

DAOs & Communities

On-chain verified links are ideal for DAOs sharing governance proposals, Discord communities, and Web3 projects needing trusted URLs.

Go-To-Market strategy

- **Phase 1**: Web3-native users & DAOs
  Integrate with Discord bots, governance tools, on-chain communities
- **Phase 2**: Developer tools
  Offer durl.dev API for projects that need verified, permanent link references
- **Phase 3**: General public
  Position as the trustworthy alternative to Bit.ly for anyone who cares about link permanence

07 - VALIDATION & FEEDBACK

Validating the market

We are pre-launch, but our validation strategy is structured and underway.

Planned feedback cycles

Hedera Community

Post on Hedera Discord & forum for early feedback from ecosystem developers and power users.

DAO & Web3 Users

Approach DAO communities directly, offer free custom slugs for feedback on usability and trust.

General Public Beta

Open beta at durl.dev, collect qualitative feedback via embedded feedback widget.

Developer Feedback

Share SDK integration guide with Web3 developers; gather friction points and feature requests.

Key metrics we will track

- **Links created / week**
- **Resolution requests / day**
- **Unique wallet addresses**
- **HCS messages / week**
- **Custom vs random link ratio**
- **Churn (links abandoned vs active)**

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08 - FUTURE ROADMAP

What's next?

The MVP is the foundation. Here is what we build on top of it.

Now - MVP Live	Phase 2 Q3 2026	Phase 3 Q4 2026	Phase 4 2027
• Random & custom link creation on Hedera	• Random & custom link creation on Hedera	• Random & custom link creation on Hedera	• Random & custom link creation on Hedera
• HCS audit log for every link event	• HCS audit log for every link event	• HCS audit log for every link event	• HCS audit log for every link event
• MetaMask + ethers.js frontend at durl.dev	• MetaMask + ethers.js frontend at durl.dev	• MetaMask + ethers.js frontend at durl.dev	• MetaMask + ethers.js frontend at durl.dev
• Mirror Node history endpoint	• Mirror Node history endpoint	• Mirror Node history endpoint	• Mirror Node history endpoint